

Sanjay J K

Software Engineer — AI Systems Developer
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SUMMARY

Computer Science student at VIT (CGPA 9.34) building AI systems and scalable full-stack applications. Experience across **LLM pipelines, RAG systems, multi-agent architectures, and production backend engineering**. Filed two AI patents and selected as an international research scholar (UKIERI-SPARC).

SKILLS

AI / ML: LLM Pipelines, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, PyTorch, Scikit-learn, Model Deployment

Systems & Backend: Event-Driven Architecture, REST APIs, Asynchronous Job Processing, Data Structures & Algorithms, Docker, Redis, PostgreSQL, MongoDB, SQL, ChromaDB

Languages: Python, TypeScript, JavaScript, C++, Java, C

Frameworks & Web: React, Next.js, FastAPI, Node.js, Express.js, TailwindCSS

Tools: Git, GitHub, LangChain, LangGraph, Langfuse, Ollama, Docker Compose

PROJECTS

LLM Knowledge Assistant (Production RAG System) Python, LangChain, ChromaDB

- Built a **Retrieval-Augmented Generation system** ingesting PDFs, Markdown, and web content, using hybrid vector + BM25 retrieval with reciprocal rank fusion.
- Added cross-encoder reranking and evaluation pipelines with **Langfuse observability**.

InterviewAI — Real-Time Voice AI Interviewer FastAPI, React, WebSockets, Ollama

- Built a real-time voice AI interviewer (Deepgram ASR, Ollama LLMs, ElevenLabs TTS) achieving **sub-1.2s end-to-end latency** via an event-driven, full-duplex WebSocket streaming architecture.
- Implemented session telemetry dashboards tracking ASR/LLM/TTS latency, with automated feedback generation and exportable interview reports.

NextUp — AI-Powered Placement Intelligence Platform Next.js, FastAPI, PostgreSQL, Redis

- Built an **AI-powered email parsing pipeline** that structures Gmail placement data into a normalized 20-table schema, eliminating manual tracking across hundreds of emails per season.
- Architected an **event-driven ingestion pipeline** (Gmail → parser → validation → workspace → notifications) with pending-event reconciliation for out-of-order, multi-role emails.
- Designed a **zero-knowledge encrypted vault** for student data and a resume generation pipeline combining structured Gemini API outputs with deterministic LaTeX rendering.

Neo4j GraphReasoner — NLP Knowledge Graph Neo4j, Node.js, Ollama

- Built an interactive research graph with multi-hop Cypher reasoning and physics-based visualization.
- Developed an LLM-powered query system translating natural language to Cypher, with full observability (Langfuse, tracing, alerts).

PATENTS

- Published (2026, Indian Patent Office):** Pan-Chronological Vision–Language Transformer for Cross-Era Tamil Script Decipherment — a VLM system for reading historical Tamil inscriptions across centuries of script drift.
- Published (2026, Indian Patent Office):** Personalized AI System for Color Vision Deficiency — an adaptive recommendation engine using perceptual gamut modeling to assist colorblind users.

EXPERIENCE

International Research Scholar — UKIERI-SPARC Program, Heriot-Watt University 2025

- Selected as **1 of 9 students nationally** for an international AI/ML research collaboration between VIT and Heriot-Watt University.

Frontend Developer Intern — Valsco Technology (Remote) Jun 2024 – Sep 2024

- Scaled **NoteSwap** to **200+ active student users** by architecting a component-based React frontend with reusable UI modules and REST API integration.
- Cut page load time by **20%**, measured via frontend performance audits, by implementing lazy loading and component-level render optimization.
- Improved cross-device usability by designing a responsive component library adopted across the full desktop and mobile interface.

EDUCATION

Vellore Institute of Technology (VIT) — B.Tech Computer Science 2023 – 2027

CGPA: **9.34**

CBSE High School

Score: **93.5%**

RECOGNITION

- GATE CSE 2026: AIR 1603** (Score: **670**).
- UKIERI Research Scholar (2025) – Heriot-Watt University, UK:** Selected among 9 students nationally.
- Judges' Choice Award – Sankalp Innovation Challenge 2026 (MNNIT Allahabad):** Built JanVedha AI, a civic decision intelligence platform.

HACKATHONS & LEADERSHIP

- Finalist – Google Cloud Agentic AI Hackathon 2025; Finalist – India Innovates 2026 (Bharat Mandapam); Top 3 – DeepShiva Open Source LLM Hackathon.
- Board Member (Design Lead), AEE VIT & Senior Core Member, CodeChef VIT / TAM-VIT.